

Pearson Edexcel Award

Wednesday 24 June 2020

Morning (Time: 3 hours)

Paper Reference **9811/01**

**Advanced Extension Award
Mathematics**

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Pearson

3. (a) (i) Write down the binomial series expansion of

$$\left(1 + \frac{2}{n}\right)^n \quad n \in \mathbb{N}, n > 2$$

in powers of $\left(\frac{2}{n}\right)$ up to and including the term in $\left(\frac{2}{n}\right)^3$

(ii) Hence prove that, for $n \in \mathbb{N}, n \geq 3$

$$\left(1 + \frac{2}{n}\right)^n \geq \frac{19}{3} - \frac{6}{n} \quad (3)$$

(b) Use the binomial series expansion of $\left(1 - \frac{x}{4}\right)^{\frac{1}{2}}$ to show that $\sqrt{3} < \frac{7}{4}$ (4)

$$f(x) = \left(1 + \frac{2}{x}\right)^x - 3^{\frac{x}{6}} \quad x \in \mathbb{R}, x > 0$$

Given that the function $f(x)$ is continuous and that $\sqrt[6]{3} > \frac{6}{5}$

(c) prove that $f(x) = 0$ has a root in the interval $[9, 10]$ (5)
(+S1)

(Total for Question 3 is 13 marks)

4.

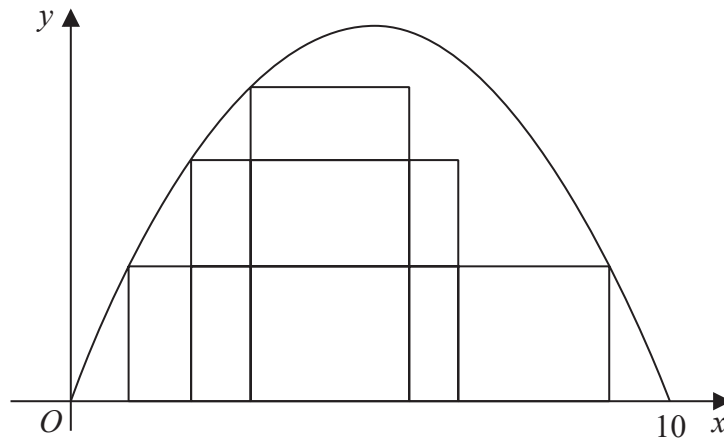


Figure 2

Figure 2 shows a sketch of the parabola with equation $y = \frac{1}{2}x(10 - x)$, $0 \leq x \leq 10$

This question concerns rectangles that lie under the parabola in the first quadrant. The bottom edge of each rectangle lies along the x -axis and the top left vertex lies on the parabola. Some examples are shown in Figure 2.

Let the x coordinate of the top left vertex be a .

(a) Explain why the width, w , of such a rectangle must satisfy $w \leq 10 - 2a$ (2)

(b) Find the value of a that gives the maximum area for such a rectangle. (5)

Given that the rectangle must be a square,

(c) find the value of a that gives the maximum area for such a square. (3)

Given that the area of the rectangles is fixed as 36

(d) find the range of possible values for a . (6)

(+S1)

(Total for Question 4 is 17 marks)

5. (a) The box below shows a student's attempt to prove the following identity for $a > b > 0$

$$\arctan a - \arctan b \equiv \arctan \frac{a - b}{1 + ab}$$

Let $x = \arctan a$ and $y = \arctan b$, so that $a = \tan x$ and $b = \tan y$

$$\text{So } \tan(\arctan a - \arctan b) \equiv \tan(x - y)$$

$$\begin{aligned} &\equiv \frac{\tan x - \tan y}{1 - \tan^2(xy)} \\ &\equiv \frac{a - b}{1 - (ab)^2} \\ &\equiv \frac{a - ab + ab - b}{(1 - ab)(1 + ab)} \\ &\equiv \frac{a(1 - ab) - b(1 - ab)}{(1 - ab)(1 + ab)} \\ &\equiv \frac{a - b}{1 + ab} \end{aligned}$$

Taking $\arctan$ of both sides gives $\arctan a - \arctan b \equiv \arctan \frac{a - b}{1 + ab}$ as required.

There are three errors in the proof where the working does not follow from the previous line.

- (i) Describe these three errors.

(3)

- (ii) Write out a correct proof of the identity.

(2)

- (b) [In this question take g to be 9.8 m s^{-2}]

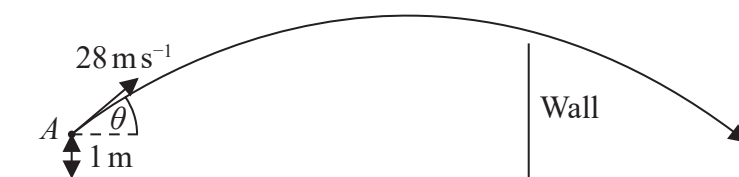


Diagram
not to scale

Figure 3

Balls are projected, one after another, from a point, A , one metre above horizontal ground. Each ball travels in a vertical plane towards a 6 metre high vertical wall of negligible thickness, which is a horizontal distance of $10\sqrt{2}$ metres from A .

The balls are modelled as particles and it is assumed that there is no air resistance.

Each ball is projected with an initial speed of 28 m s^{-1} and at a random angle θ to the horizontal, where $0 < \theta < 90^\circ$

Question 5 continued

Given that a ball will pass over the wall precisely when $\alpha \leq \theta \leq \beta$

(i) find, in degrees, the angle $\beta - \alpha$ (10)

(ii) Deduce that the probability that a particular ball will pass over the wall is $\frac{2}{3}$ (1)

(iii) Hence find the probability that exactly 2 of the first 10 balls projected pass over the wall.

You should give your answer in the form $\frac{P}{Q^k}$ where P , Q and k are integers and P is not a multiple of Q .

(3)

(iv) Explain whether taking air resistance into account would increase or decrease the probability in (b)(iii).

(1)

(+S2)

(Total for Question 5 is 22 marks)

6. (a) Given that f is a function such that the integrals exist,

(i) use the substitution $u = a - x$ to show that

$$\int_0^a f(x) dx = \int_0^a f(a - x) dx \quad (2)$$

(ii) Hence use symmetry of $f(\sin x)$ on the interval $[0, \pi]$ to show that

$$\int_0^\pi x f(\sin x) dx = \pi \int_0^{\frac{\pi}{2}} f(\sin x) dx \quad (4)$$

(b) Use the result of (a)(i) to show that

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx$$

is independent of n , and find the value of this integral.

(4)

(c) (i) Prove that

$$\frac{\cos x}{1 + \cos x} \equiv 1 - \frac{1}{2} \sec^2 \left(\frac{x}{2} \right)$$

(ii) Hence use the results from (a) to find

$$\int_0^\pi \frac{x \sin x}{1 + \sin x} dx \quad (7)$$

(d) Find

$$\int_0^\pi \frac{x \sin^4 x}{\sin^4 x + \cos^4 x} dx \quad (4)$$

(+S2)

(Total for Question 6 is 23 marks)

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